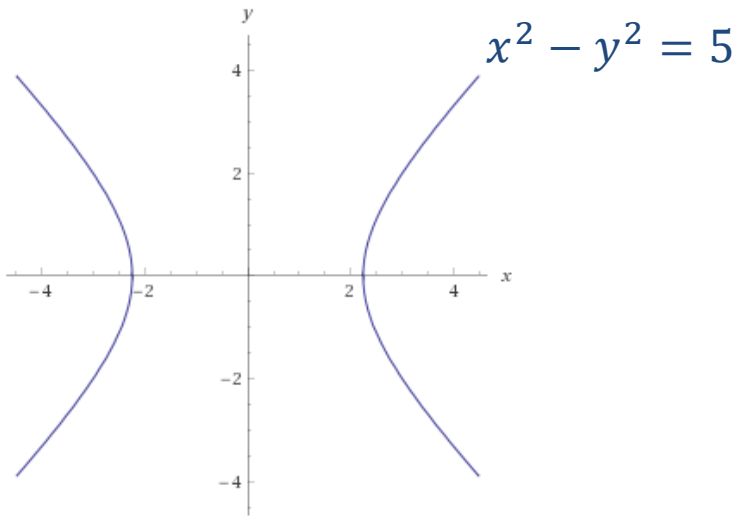
P2 Chapter 8: Parametric Equations

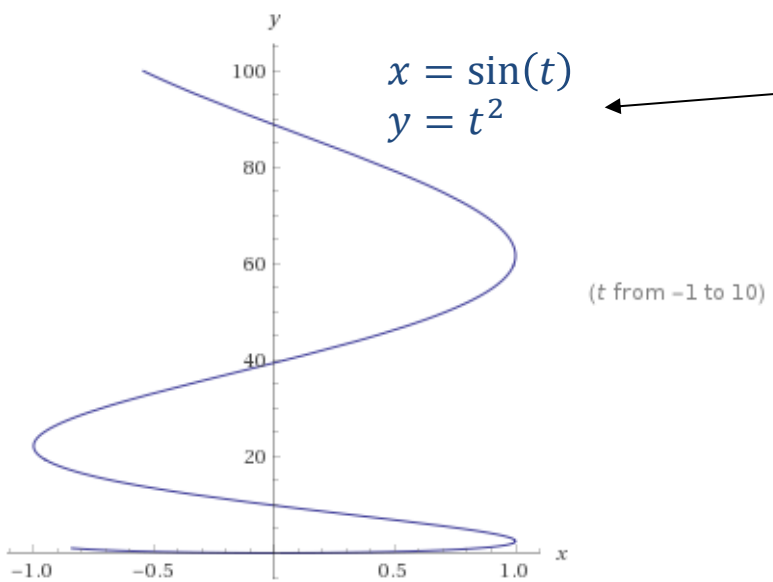
Using Parameters

What are they and what is the point?



Typically, with two variables x and y, we can relate the two by **a single equation involving just x and y**.
This is known as a **<u>Cartesian equation</u>**.

The line shows all points (x, y) which satisfy the Cartesian equation.



However, in Mechanics for example, we might want each of the x and y values to be some function of time t, as per this example.

This would allow us to express the position of a particle at time t as the vector:

$$\begin{pmatrix} \sin t \\ t^2 \end{pmatrix}$$

These are known as **parametric equations**, because **each of x and y are defined in terms of some other variable, known as the parameter** (in this case t).

Converting parametric to Cartesian

How could we convert these parametric equations into a single Cartesian one?

$$x = 2t, \qquad y = t^2, \qquad -3 < t < 3$$

Use substitution to eliminate the parameter.

$$\boldsymbol{t = \frac{x}{2}}$$

$$\boldsymbol{y = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4}}$$

What is the domain of the function?

✎ If $x = p(t)$ and $y = q(t)$ can be written as $y = f(x)$, then the domain of f is the range of p...

If $-3 < t < 3$ then $-6 < x < 6$. ← Use $x = 2t$ for each of the bounds.

✎ and the range of f is the range of q.

If $-3 < t < 3$ then $0 \leq y < 9$. ← This one is harder. $(-3)^2 = 9$ and $3^2 = 9$. However, the minimum value is clearly $0^2 = 0$.

Further Example

[Textbook] A curve has the parameter equations

$$x = \ln(t+3), \qquad y = \frac{1}{t+5}, \qquad t > -2$$

a) Find a Cartesian equation of the curve of the form $y = f(x)$, $x > k$, where k is a constant to be found.

b) Write down the range of $f(x)$.

? a

? b

A common strategy for domain/range questions is to consider what happens are the boundary value (in this case -2), then since $t > -2$, consider what happens as t increases.

Further Example

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a) Find a Cartesian equation of the curve of the form $y = f(x)$, $x > k$, where k is a constant to be found.

b) Write down the range of $f(x)$.

$$x = \ln(t+3)$$
$$e^x = t + 3$$
$$e^x - 3 = t$$
$$\therefore y = \frac{1}{e^x - 3 + 5} = \frac{1}{e^x + 2}$$

A common strategy for domain/range questions is to consider what happens are the boundary value (in this case -2), then since $t > -2$, consider what happens as t increases.

When $t = -2$, $x = \ln(-2+3) = \ln(1) = 0$

As t increases, $\ln(t+3)$ increases, therefore $x > 0$.

$\therefore k = 0$

b) When $t = -2$, $y = \frac{1}{-2+5} = \frac{1}{3}$

As t increases, $\frac{1}{t+5}$ decreases, therefore $y < \frac{1}{3}$

Test Your Understanding

Edexcel C4 Jan 2008 Q7

The curve C has parametric equations

$$x = \ln(t+2), \quad y = \frac{1}{(t+1)}, \quad t > -1.$$

(c) Find a cartesian equation of the curve C, in the form $y = f(x)$. **(4)**

?

Edexcel C4 Jan 2011

6. The curve C has parametric equations

$$x = \ln t, \qquad y = t^2 - 2, \qquad t > 0.$$

(b) a cartesian equation of C.

(3)

?

Test Your Understanding

Edexcel C4 Jan 2008 Q7

The curve C has parametric equations

$$x = \ln(t+2), \quad y = \frac{1}{(t+1)}, \quad t > -1.$$

(c) Find a cartesian equation of the curve C, in the form $y = f(x)$. **(4)**

$x = \ln(t+2), \quad y = \frac{1}{t+1}$

$e^x = t+2 \Rightarrow t = e^x - 2$

$y = \frac{1}{e^x - 2 + 1} \Rightarrow y = \frac{1}{e^x - 1}$

Attempt to make $t = \ldots$ the subject	M1
giving $t = e^x - 2$	A1
Eliminates t by substituting in y	dM1
giving $y = \frac{1}{e^x - 1}$	A1

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6. The curve C has parametric equations

$$x = \ln t, \qquad y = t^2 - 2, \qquad t > 0.$$

(b) a cartesian equation of C.

(3)

$t = e^x$

$y = (e^x)^2 - 2$

$= e^{2x} - 2$

Exercise 8.1

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Homework Exercise

1 Find a Cartesian equation for each of these parametric equations, giving your answer in the form $y = f(x)$. In each case find the domain and range of $f(x)$.

a $x = t - 2, \quad y = t^2 + 1, \quad -4 \leqslant t \leqslant 4$

b $x = 5 - t, \quad y = t^2 - 1, \quad t \in \mathbb{R}$

c $x = \frac{1}{t}, \quad y = 3 - t, \quad t \neq 0$

Notation If the domain of t is given as $t \neq 0$, this implies that t can take any value in $\mathbb{R}$ other than 0.

d $x = 2t + 1, \quad y = \frac{1}{t}, \quad t > 0$

e $x = \frac{1}{t - 2}, \quad y = t^2, \quad t > 2$

f $x = \frac{1}{t + 1}, \quad y = \frac{1}{t - 2}, \quad t > 2$

2 For each of these parametric curves:

i find a Cartesian equation for the curve in the form $y = f(x)$ giving the domain on which the curve is defined

ii find the range of $f(x)$.

a $x = 2\ln(5 - t), \quad y = t^2 - 5, \quad t < 4$

b $x = \ln(t + 3), \quad y = \frac{1}{t + 5}, \quad t > -2$

c $x = e^t, \quad y = e^{3t}, \quad t \in \mathbb{R}$

3 A curve C is defined by the parametric equations $x = \sqrt{t}, \quad y = t(9 - t), \quad 0 \leqslant t \leqslant 5$.

a Find a Cartesian equation of the curve in the form $y = f(x)$, and determine the domain and range of $f(x)$.

b Sketch C showing clearly the coordinates of any turning points, endpoints and intersections with the coordinate axes.

Problem-solving

$y = t(9 - t)$ is a quadratic with a negative t^2 term and roots at $t = 0$ and $t = 9$. It will take its **maximum** value when $t = 4.5$.

Homework Exercise

4 For each of the following parametric curves:

i find a Cartesian equation for the curve in the form $y = f(x)$

ii find the domain and range of $f(x)$

iii sketch the curve within the given domain of t.

a $x = 2t^2 - 3, \quad y = 9 - t^2, \quad t > 0$

b $x = 3t - 1, \quad y = (t - 1)(t + 2), \quad -4 < t < 4$

c $x = t + 1, \quad y = \dfrac{1}{t - 1}, \quad t \in \mathbb{R}, \quad t \neq 1$

d $x = \sqrt{t} - 1, \quad y = 3\sqrt{t}, \quad t > 0$

e $x = \ln(4 - t), \quad y = t - 2, \quad t < 3$

5 The curves C_1 and C_2 are defined by the following parametric equations.

C_1: $x = 1 + 2t, \quad y = 2 + 3t \quad 2 < t < 5$

C_2: $x = \dfrac{1}{2t - 3}, \quad y = \dfrac{t}{2t - 3} \quad 2 < t < 3$

a Show that both curves are segments of the same straight line.

b Find the length of each line segment.

Notation Straight lines and line segments can be referred to as 'curves' in coordinate geometry.

6 A curve C has parametric equations

$$x = \frac{3}{t} + 2, \quad y = 2t - 3 - t^2, \quad t \in \mathbb{R}, \quad t \neq 0$$

a Determine the ranges of x and y in the given domain of t. **(3 marks)**

b Show that the Cartesian equation of C can be written in the form

$$y = \frac{A(x^2 + bx + c)}{(x - 2)^2}$$

where A, b and c are integers to be determined. **(3 marks)**

Homework Exercise

7 A curve has parametric equations

$$x = \ln(t+3), \quad y = \frac{1}{t+5}, \quad t > -2$$

a Show that a Cartesian equation of this curve is $y = f(x)$, $x > k$ where k is a constant to be found.

b Write down the range of $f(x)$.

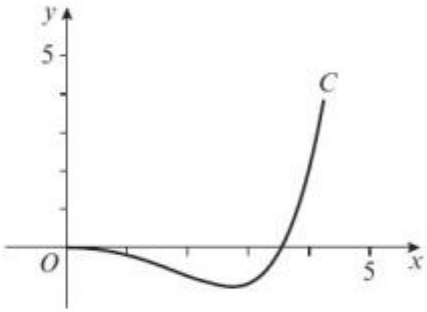
8 A diagram shows a curve C with parametric equations

$$x = 3\sqrt{t}, \quad y = t^3 - 2t, \quad 0 \leqslant t \leqslant 2$$

a Find a Cartesian equation of the curve in the form $y = f(x)$, and state the domain of $f(x)$. **(3 marks)**

b Show that $\frac{dy}{dt} = 0$ when $t = \sqrt{\frac{2}{3}}$ **(3 marks)**

c Hence determine the range of $f(x)$. **(2 marks)**



9 A curve C has parametric equations

$$x = t^3 - t, \quad y = 4 - t^2, \quad t \in \mathbb{R}$$

a Show that the Cartesian equation of C can be written in the form

$$x^2 = (a - y)(b - y)^2$$

where a and b are integers to be determined. **(3 marks)**

b Write down the maximum value of the y-coordinate for any point on this curve. **(2 marks)**

Homework Exercise

Challenge

A curve C has parametric equations

$$x = \frac{1 - t^2}{1 + t^2}, \quad y = \frac{2t}{1 + t^2}, \quad t \in \mathbb{R}$$

a Show that a Cartesian equation for this curve is $x^2 + y^2 = 1$.

b Hence describe C.

Homework Answers

1 **a** $y = (x+2)^2 + 1, -6 \leqslant x \leqslant 2,$ $\quad 1 \leqslant y \leqslant 17$

b $y = (5-x)^2 - 1, x \in \mathbb{R}$ $\quad y \geqslant -1$

c $y = 3 - \dfrac{1}{x}, x \neq 0,$ $\quad y \neq 3$

d $y = \dfrac{2}{x-1}, x > 1,$ $\quad y > 0$

e $y = \left(\dfrac{1+2x}{x}\right)^2, x > 0,$ $\quad y > 4$

f $y = \dfrac{x}{1-3x}, 0 < x < \dfrac{1}{3},$ $\quad y > 0$

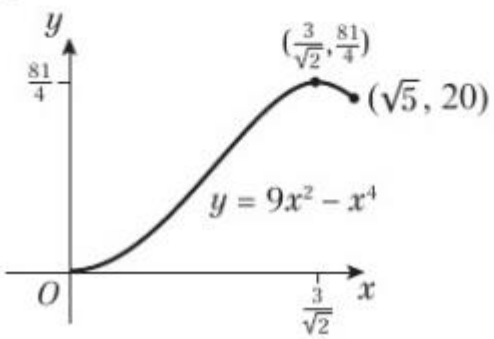
2 **a** **i** $y = 20 - 10e^{\frac{1}{2}x} + e^x, x > 0$ $\quad$ **ii** $y \geqslant -5$

b **i** $y = \dfrac{1}{e^x + 2}, x > 0$ $\quad$ **ii** $0 < y < \frac{1}{3}$

c **i** $y = x^3, x > 0$ $\quad$ **ii** $y > 0$

3 **a** $y = 9x^2 - x^4,$ $\quad 0 \leqslant x \leqslant \sqrt{5},$ $\quad 0 \leqslant y \leqslant \frac{81}{4}$

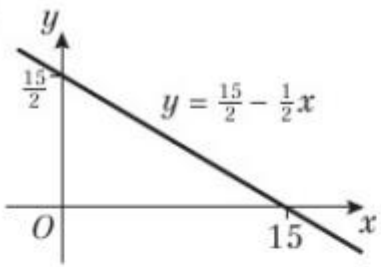
b



Homework Answers

4 **a** **i** $y = \frac{15}{2} - \frac{1}{2}x$ **ii** $x > -3, y < 9$

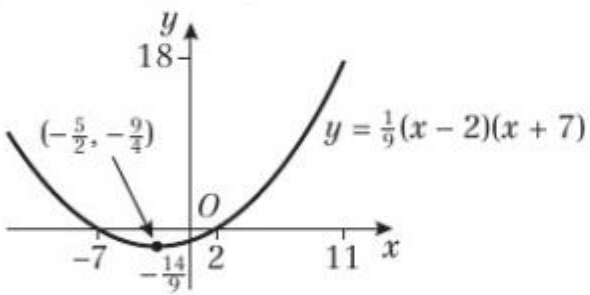
iii



b **i** $y = \frac{1}{9}(x - 2)(x + 7)$

ii $-13 < x < 11, -\frac{9}{4} < y < 18$

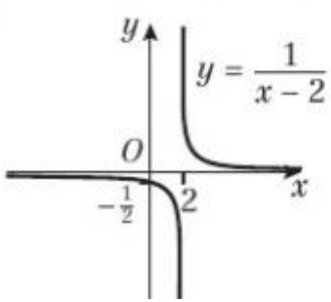
iii



c **i** $y = \dfrac{1}{x - 2}$

ii $x \in \mathbb{R}, x \neq 2, y \in \mathbb{R}, y \neq 0$

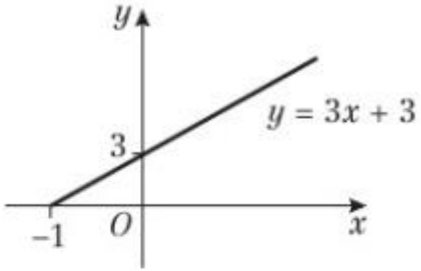
iii



d **i** $y = 3x + 3$

ii $x > -1, y > 0$

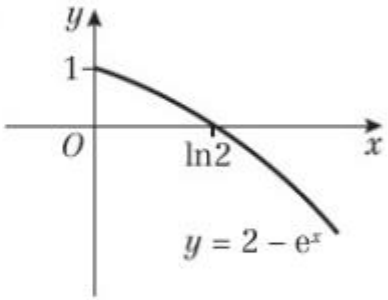
iii



e **i** $y = 2 - e^x$

ii $x > 0, y < 1$

iii



Homework Answers

5 **a** C_1: $x = 1 + 2t$, $t = \dfrac{x-1}{2}$

Sub t into $y = 2 + 3t$:

$$y = 2 + 3\left(\frac{x-1}{2}\right) = 2 + \frac{3}{2}x - \frac{3}{2} = \frac{3}{2}x + \frac{1}{2}$$

C_2: $x = \dfrac{1}{2t-3}$, $t = \dfrac{1+3x}{2x}$

Sub into $y = \dfrac{t}{2t-3}$.

$$y = \frac{\dfrac{1+3x}{2x}}{2\left(\dfrac{1+3x}{2x}\right) - 3} = \frac{\dfrac{1+3x}{2x}}{\dfrac{1}{x}} = \frac{1+3x}{2} = \frac{3}{2}x + \frac{1}{2}$$

Therefore C_1 and C_2 represent a segment of the same straight line.

b Length of $C_1 = 3\sqrt{13}$, length of $C_2 = \dfrac{\sqrt{13}}{3}$

6 **a** $x \neq 2$; $y \leqslant -2$, $y \neq -3$

b $x = \dfrac{3}{t} + 2$, $t = \dfrac{3}{x-2}$

Sub into $y = 2t - 3 - t^2$

$$y = 2\left(\frac{3}{x-2}\right) - 3 - \left(\frac{3}{x-2}\right)^2 = \frac{6}{x-2} - 3 - \frac{9}{(x-2)^2}$$

$$= \frac{6(x-2) - 3(x-2)^2 - 9}{(x-2)^2}$$

$$= \frac{6x - 12 - 3x^2 + 12x - 12 - 9}{(x-2)^2} = \frac{-3x^2 + 18x - 33}{(x-2)^2}$$

$$= \frac{-3(x^2 - 6x + 11)}{(x-2)^2} \text{ so } A = -3,\ b = -6,\ c = 11$$

Homework Answers

7 **a** $x = \ln(t+3)$ $\quad t = e^x - 3$ $\quad$ Sub into $y = \dfrac{1}{t+5}$

$$y = \frac{1}{e^x - 3 + 5} = \frac{1}{e^x + 2}, \quad x > 0$$

b $0 < y < \dfrac{1}{3}$

8 **a** $y = \dfrac{x^6}{729} - \dfrac{2x^2}{9}, 0 \leqslant x \leqslant 3\sqrt{2}$

b $y = t^3 - 2t, \dfrac{dy}{dt} = 3t^2 - 2$

$0 = 3t^2 - 2 \qquad t^2 = \dfrac{2}{3} \qquad t = \sqrt{\dfrac{2}{3}}$

c $-\dfrac{4\sqrt{6}}{9} \leqslant f(x) \leqslant 4$

9 **a** $y = 4 - t^2 \Rightarrow t = \sqrt{4-y}$

Sub into $x = t^3 - t = t(t^2 - 1)$

$x = \sqrt{4-y}(4 - y - 1) = \sqrt{4-y}(3-y)$

$x^2 = (4-y)(3-y)^2$

$a = 4, b = 3$

b Max y is 4

Challenge

a $x^2 = \dfrac{(1-t^2)^2}{(1+t^2)^2}, y^2 = \dfrac{4t^2}{(1+t^2)^2}$

$$x^2 + y^2 = \frac{(1-t^2)^2}{(1+t^2)^2} + \frac{4t^2}{(1+t^2)^2} = \frac{1 - 2t^2 + t^4}{(1+t^2)^2} + \frac{4t^2}{(1+t^2)^2}$$

$$= \frac{1 + 2t^2 + t^4}{(1+t^2)^2} = \frac{(1+t^2)^2}{(1+t^2)^2} = 1$$

So $x^2 + y^2 = 1$

b Circle, centre (0,0), radius 1.